TEXORA · LECTURE SHEET

Operations and Supply Chain Management in the Textile and Apparel Industry

From cotton field to retail shelf: operations management inside the factory, the structure of Bangladesh's RMG chain, push versus pull, the bullwhip effect, lead time, inventory control with the EOQ model, and vendor selection.

TE-3109 · SUPPLY CHAIN MANAGEMENT IN TEXTILES · TOPIC 5 · LEVEL 3 TERM 1

1. Operations management — the factory's core

Operations management is the management of the transformation process that converts inputs — fibre, yarn, fabric, chemicals, labour, machine time, information — into outputs of greater value: yarn, fabric, finished garments. Every earlier topic meets here: structure (Topic 2) arranges the transformation, people (Topic 3) perform it, technology (Topic 4) equips it. The operations manager's permanent concerns are the five performance objectives: **quality** (make it right), **speed** (make it fast), **dependability** (deliver on time), **flexibility** (change what, how much and when) and **cost** (make it cheaply enough to profit). Buyers rank these openly: a garment exporter is judged on quality and dependability first, because a failed audit or a missed vessel costs the buyer a season — cost decides only among suppliers who clear those two bars.

Key operations decisions in a textile firm include process choice (batch dyeing versus continuous ranges; line versus modular sewing), capacity planning (how many lines, looms or ring frames the order book justifies, and how seasonal peaks are covered), layout and materials handling (keeping fabric, work-in-progress and finished goods moving in one direction with minimal backtracking), production planning and control (order scheduling, loading, sequencing and progress chasing), quality management, maintenance management, and inventory management (Section 6). Operations decisions are where strategy becomes physical: a firm's promise of a 45-day lead time is either true in its capacity and inventory plan, or it is fiction.

2. Supply chain management — concepts

A **supply chain** is the network of organisations, people, activities, information and resources involved in supplying a product from raw material to final consumer. **Supply chain management (SCM)** is the

integration and coordination of that network — the flows of **materials** (forward, supplier to customer), **information** (both ways: orders forward, forecasts and demand signals backward) and **funds** (backward, customer to supplier) — so that the right product reaches the right place at the right time, in the right quantity and quality, at the lowest total cost of the whole chain rather than of any single firm in it.

Three distinctions matter. First, **logistics** — transport, warehousing and distribution — is a part of SCM, not the whole: SCM also covers sourcing strategy, supplier relationships, production coordination and demand planning. Second, supply chains compete with supply chains: a Bangladeshi exporter's real rival is not the factory across the road but the Vietnamese or Ethiopian chain that delivers the same shirt to the same retailer faster and more reliably. Third, integration is a spectrum: from arm's-length buying (each order negotiated alone, information hoarded) through coordination (shared schedules) to genuine collaboration (shared forecasts, joint planning, long-term partnerships). The direction of the whole industry is toward fewer, deeper supplier relationships — buyers concentrate orders in factories they trust and develop — because coordination failures, not factory effort, are where most lead time and cost hide.

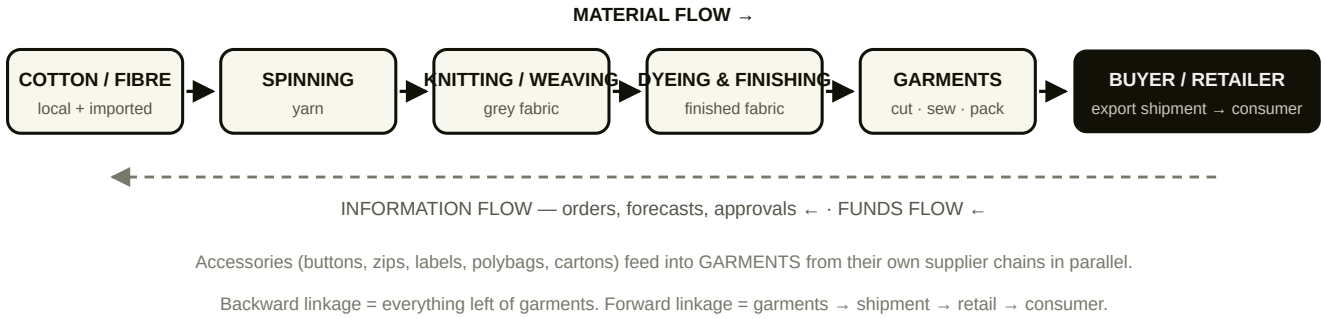


Figure 1 — The textile and apparel supply chain. Material flows forward from fibre to consumer; information and funds flow back. Every arrow is a place where delay, cost or quality failure can enter.

3. The structure of the Bangladesh RMG supply chain

Bangladesh's ready-made garment industry is the clearest living example of a supply chain in the national economy. Its structure is best read through **linkages**. **Backward linkage** industries supply the garment factory: spinning mills (yarn), knitting and weaving mills (fabric), dyeing and finishing units, and the accessories (trim) industries — buttons, zippers, sewing thread, labels, hangers, polybags and cartons. The knitwear sub-sector enjoys strong backward linkage: local spinning, knitting and dyeing can supply most of its fabric needs within the country, which shortens lead time and adds local value. The woven sub-sector is historically more dependent on **imported fabric**, mainly woven cloth brought in against back-to-back letters of credit, so its chain carries an international shipping leg — and its lead-

time risk — inside almost every order. **Forward linkage** runs from the factory gate outward: freight forwarders and shipping lines, the buyer's distribution centres, retail stores and finally consumers, together with the banking and inspection services that move documents and verify quality along the way.

The chain is coordinated commercially by **merchandisers** and **buying houses / liaison offices**: the buyer places the order with specifications and a delivery date; the factory's merchandiser converts it into bookings of yarn or fabric, trims and capacity; production planning sequences dyeing, cutting and sewing; inspection agencies verify quality before shipment; and documents flow through banks under the letter-of-credit system that finances the whole cycle. Two structural lessons follow. First, the garment factory — the visible centre of the industry — captures only part of the chain's value; strength in the backward links (yarn, fabric, dyeing quality and speed) decides how much value and how much lead-time control stays in the country. Second, every hand-off is an information event: most delays blamed on "production" are born earlier, in late approvals, late fabric or late trims — chain problems wearing a factory costume.

4. Push versus pull, and the bullwhip effect

FEATURE	PUSH SYSTEM	PULL SYSTEM
Trigger	Forecast: produce in advance of demand and push stock forward	Actual demand: each stage produces only what the next stage has consumed or ordered
Inventory	High — stock held at every stage as insurance	Low — stock kept minimal, replenished frequently in small lots
Lead-time behaviour	Long planned cycles; slow response to fashion change	Short cycles; fast response, demands reliable suppliers and fast information
Risk	Obsolescence and markdowns when the forecast is wrong	Stock-outs and production stops when supply falters
Textile face	Spinning and grey-fabric production to forecast; basic yarns stocked by count	Dyeing and garment making strictly against confirmed orders; quick-response programmes with retailers

Real chains mix both: push where demand is stable and production is capital-heavy (spinning standard counts), pull where variety explodes and fashion rules (dyeing colours, garment styles). The point where the chain switches from push to pull — the **decoupling point** — is a strategic choice: holding grey fabric (push up to fabric, pull from dyeing onward) is the classic textile compromise, because grey fabric is generic while dyed fabric is committed to one buyer's colour.

The **bullwhip effect** is the amplification of demand variation as orders move upstream: a small change in retail sales becomes a larger swing in the retailer's orders, a still larger swing in the garment factory's fabric bookings, and a violent swing at the spinning mill. Its causes are demand-forecast updating at every stage, order batching (buying monthly instead of weekly), price fluctuations that invite forward buying, and rationing games when supply is short. Its consequences are the industry's familiar diseases: alternating yarn gluts and famines, overtime followed by idle machines, and warehouses full of the wrong counts. The cures are informational: share real demand data instead of orders alone, order little and often, keep prices stable, and build collaborative planning with key partners — in short, manage the chain as one system (Section 2) instead of a line of strangers.

5. Lead time

Lead time is the elapsed time between placing an order (or starting a process) and receiving the goods (or completing it). In export garments the **order lead time** — from order confirmation to shipment — is a headline competitive number, decomposed into fabric lead time (yarn booking, knitting or weaving, dyeing and finishing, plus import time where fabric is foreign), trims lead time, sampling and approval time, production time (cutting, sewing, finishing) and shipment documentation time. Buyers continuously press for shorter lead times because fashion demand is forecast poorly far in advance: every week removed from the chain is a week of demand the buyer can actually see before committing. A factory shortens lead time not by sewing faster — sewing is a small part of the total — but by attacking the waits: parallel rather than serial approvals, standard fabrics held as grey stock, pre-booked dyeing capacity, local trims sourcing, and honest capacity booking that refuses orders the chain cannot serve. Internally, **manufacturing lead time** for a lot is mostly waiting (in queues before dyeing, in bundles before sewing); work-in-progress and lead time rise and fall together, which is why lean factories obsess over small transfer batches and visible queues.

6. Inventory — types, costs and control

Inventory is stock held to decouple stages that produce and consume at different rates and times. In a textile firm it appears as **raw material inventory** (cotton, yarn, grey fabric, dyes and chemicals, trims), **work-in-progress (WIP)** (fabric in dyeing, cut panels, garments on lines), **finished goods inventory** (packed garments awaiting shipment, yarn cones in the godown) and **maintenance and spare-parts inventory**. Inventory is simultaneously a service and a cost: it protects production and delivery against uncertainty, and it locks up working capital, warehouse space and management attention, while risking deterioration (yarn moisture and strength changes), shade and fashion obsolescence, and hidden quality decay. Inventory management is the discipline of holding the least stock that still delivers the promised service.

The two cost families that every inventory decision balances are **ordering (procurement) costs** — the cost of placing and receiving one order: paperwork, inspection, transport arrangements, payment processing — and **holding (carrying) costs** — the cost of keeping one unit in stock for a year: capital interest, storage, insurance, handling, deterioration and obsolescence, usually expressed as a rate per unit per year. Order large and seldom, and ordering cost falls while holding cost rises; order small and often, and the reverse. **ABC analysis** focuses control where the money is: a few **A items** (high annual consumption value — main yarns, main fabrics, key dyes) are controlled tightly with records and review; many **C items** (low value — common trims and consumables) are controlled simply with generous buffers; **B items** sit between. Control effort should follow value, not item count.

6.1 The economic order quantity (EOQ)

For an item consumed at a steady rate, bought at a constant price, with no shortages allowed, the **economic order quantity** is the order size that minimises total relevant cost — ordering cost plus holding cost. Let D = annual demand (units/year), S = cost per order, H = holding cost per unit per year, and Q = order quantity. Orders per year = D/Q, so annual ordering cost = (D/Q) × S. Average inventory = Q/2, so annual holding cost = (Q/2) × H. Total cost is minimised where the two are equal, giving:

$$EOQ = \sqrt{(2DS / H)}$$

The **reorder point** completes the system: with steady daily use and a known lead time, reorder when stock falls to (daily use × lead time in days), plus any safety stock held against variability.

> **VERIFIED WORKED EXAMPLE — EOQ FOR YARN PURCHASE**
>
> A knitting factory buys yarn with annual demand **D = 120,000 kg**, ordering cost **S = Tk 1,500 per order**, holding cost **H = Tk 60 per kg per year**.
>
> EOQ = √(2DS/H) = √(2×120,000×1,500/60) = √6,000,000 ≈ **2,449 kg** per order.
>
> Orders per year = 120,000 ÷ 2,449 ≈ **49 orders/year**.
>
> Annual ordering cost = 49 × Tk 1,500 ≈ **Tk 73,485**; annual holding cost = (2,449 ÷ 2) × Tk 60 ≈ **Tk 73,485** — the two are equal at the EOQ, as the model predicts. Total relevant cost ≈ **Tk 146,969** per year.
>
> Daily use = 120,000 kg ÷ 300 working days = **400 kg/day**. With a 7-day lead time, the reorder point = 400 × 7 = **2,800 kg**: place the next order when yarn stock falls to 2,800 kg (before adding any safety stock).

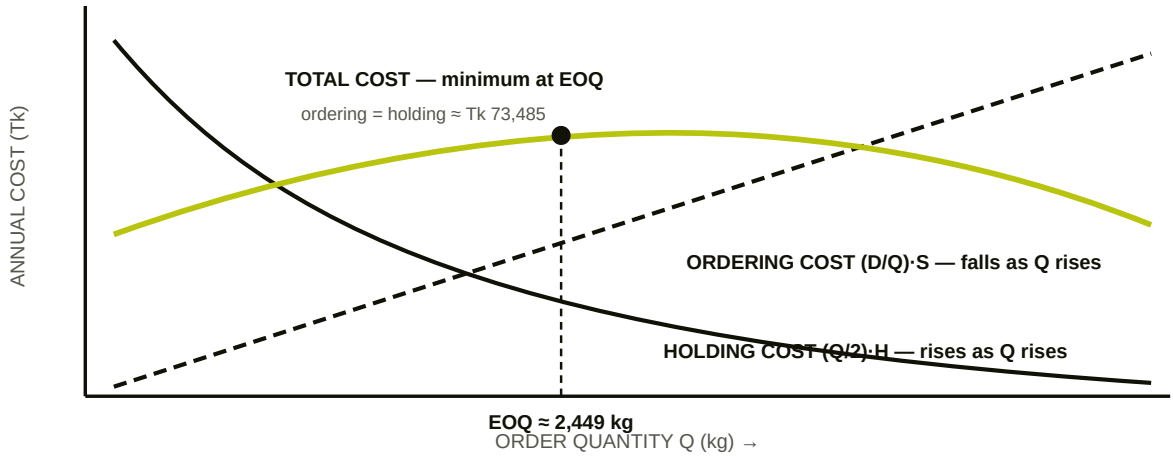


Figure 2 — The EOQ cost curves for the worked example. Ordering cost falls and holding cost rises with order size; total cost is lowest where the two curves cross — at Q ≈ 2,449 kg, where each cost is about Tk 73,485 per year.

The EOQ model's assumptions — steady demand, constant costs and price, instantaneous replenishment, no shortages — never hold perfectly, yet the model remains valuable: total cost is flat near the optimum, so moderate rounding (ordering 2,500 kg instead of 2,449 kg) costs almost nothing, while gross errors (monthly orders of 10,000 kg, or daily orders of 400 kg) are expensive and now visibly so. Where prices vary with quantity, or demand is lumpy and order-driven — as with buyer-specific dyed fabric — EOQ gives way to lot-for-lot ordering and material requirements planning against the order book.

7. Vendor (supplier) selection and relations

Vendor selection — choosing who supplies yarn, fabric, chemicals, trims and services — is a supply-chain decision with factory-level consequences: every supplier failure arrives as a production stop, a shade problem or a missed shipment wearing the factory's name. A systematic selection process runs: identify the need and specification → survey candidate suppliers → evaluate against weighted criteria → trial orders and quality verification → approve and list → review performance periodically and develop or replace. The classic evaluation criteria, weighted to the item's importance, are **quality** (consistency against specification, test results, claim history), **price and total cost** (including transport, payment terms and the cost of the supplier's failures — not invoice price alone), **delivery reliability and lead time**, **capacity and flexibility** (peak-season behaviour, small-lot willingness), **technical capability and service** (shade development support from a dye supplier, spinning consistency data from a mill), **financial stability**, and **compliance standing** — a trims supplier using child labour contaminates the buyer's chain exactly as the factory's own fault would.

CRITERION	WHAT IS EXAMINED	EVIDENCE USED

Quality	Specification conformity, consistency lot to lot, defect and claim record	Lab test reports, incoming-inspection data, trial-lot results
Total cost	Price plus transport, terms, wastage and failure cost	Quotations, landed-cost comparison, past claim costs
Delivery	Quoted versus actual lead time, on-time percentage, flexibility	Delivery history, capacity enquiry, reference buyers
Capability & service	Machinery, technical staff, development and problem-solving support	Factory visit, audit, sample development performance
Stability & compliance	Financial health, safety and social compliance, business continuity	Trade references, compliance audit reports

Selection is the start, not the end: leading firms rate vendors periodically (a **vendor rating** score per period from delivery, quality and service data), share forecasts with A-class suppliers, help key suppliers improve (supplier development), and keep a qualified second source for every critical input — because in supply chains, loyalty without alternatives is dependence, and dependence without loyalty is a bidding war. Both extremes fail; managed partnership is the middle path the whole of SCM points toward.

Glossary

Operations management — managing the input-to-output transformation process. **Supply chain** — the network from raw material to consumer carrying material, information and fund flows. **Backward linkage** — supplier industries feeding a sector (spinning, fabric, trims for garments). **Forward linkage** — the distribution path from factory to consumer. **Lead time** — elapsed time from order to receipt or completion. **Push / pull** — production triggered by forecast / by actual demand. **Bullwhip effect** — amplification of demand variation upstream in a chain. **EOQ** — economic order quantity, √(2DS/H), minimising ordering plus holding cost. **Reorder point** — stock level (daily use × lead time, plus safety stock) that triggers a new order. **ABC analysis** — inventory control priority by annual consumption value. **Vendor rating** — periodic scoring of supplier performance.

Key points

• Operations management runs the transformation; SCM integrates the whole network — and chains compete with chains.

• Bangladesh knitwear shows strong backward linkage; woven garments still carry imported-fabric lead-time risk — forward linkage runs factory → ship → retail → consumer.

• Push suits stable, capital-heavy stages; pull suits fashion stages; grey-fabric stock is the classic

decoupling point. The bullwhip effect is cured with shared information, not bigger buffers.

• Lead time is mostly waiting — across approvals, fabric and trims — not sewing speed.

• EOQ balances ordering against holding cost: for D = 120,000 kg, S = Tk 1,500, H = Tk 60, EOQ ≈ 2,449 kg, ≈ 49 orders a year, reorder point 2,800 kg at 7-day lead time.

• Vendor selection weighs quality, total cost, delivery, capability, stability and compliance — and continues as rating, development and a qualified second source.

Practice

1. Recompute the worked example with holding cost doubled to H = Tk 120/kg/year. Find the new EOQ, orders per year and total relevant cost, and explain in one sentence why EOQ falls by less than half.
2. A retailer raises orders by 5 % for one season. Trace the bullwhip effect stage by stage back to the spinning mill, naming the mechanism at each stage, then state two cures.
3. An order's 90-day lead time consists of: approvals 12 days, fabric (imported) 45 days, trims 10 days (parallel with fabric), production 18 days, inspection and shipment 5 days. Find the critical path length, then propose three changes that cut at least 20 days and state the new total.
4. Classify as raw material, WIP or finished goods: cones in the yarn godown; fabric in the dyeing queue; cut panels bundled for sewing; packed cartons awaiting inspection; spare loom reeds. Then state which class usually hides the most cash in a garment factory and why.
5. Build a weighted vendor score for two yarn suppliers: weights — quality 35 %, total cost 25 %, delivery 25 %, service/compliance 15 %. Supplier X scores 8, 7, 6, 9; Supplier Y scores 6, 9, 9, 6 (out of 10). Compute both totals and recommend, noting one non-score factor that could overturn the result.
6. For each stage — spinning standard counts, grey fabric, dyeing, garment sewing — state push or pull and justify in one line each.

Answer key

1. EOQ = √(2×120,000×1,500/120) = √3,000,000 ≈ **1,732 kg**. Orders/year = 120,000 ÷ 1,732 ≈ **69**. At the optimum ordering = holding = (1,732 ÷ 2) × 120 ≈ **Tk 103,923** each; total ≈ **Tk 207,846**. EOQ falls by the square root of the change (÷√2 ≈ 0.71), not by half, because EOQ varies with √H, not with H itself.
2. Retailer, fearing shortage, orders +10 % (forecast updating and safety padding); the garment factory, receiving lumpy larger orders, batches fabric bookings (+20 %); the dyeing/knitting unit batches yarn purchase (+35 %); the spinning mill sees a surge it reads as growth and plans capacity and cotton buying accordingly — then the correction arrives amplified in reverse. Any two cures: share

real sales/demand data along the chain; smaller, more frequent orders; stable pricing without forward-buying incentives; collaborative planning with key partners.

3. Critical path: approvals 12 + fabric 45 + production 18 + shipment 5 = **80 days** (trims at 10 days run parallel inside the fabric window and are not critical). Cuts (any three): switch to local fabric or grey-stock + local dyeing (fabric 45 → 20, saving 25); overlap approvals with fabric by approving lab dips and strike-offs against standard (saving ~6); pre-book production capacity and trims (protects the 18 days); combined effect must show a new total of 60 days or fewer, e.g., 12 + 20 + 18 + 5 = **55 days**.
4. Raw material: yarn cones, spare reeds (maintenance inventory — accept either with reasoning). WIP: fabric in dyeing queue, cut panels. Finished goods: packed cartons. In most garment factories WIP plus fabric stock hides the most cash: fabric is the largest cost component and sits longest in queues between knitting/weaving, dyeing and cutting, so value accumulates invisibly stage by stage.
5. X: 8×0.35 + 7×0.25 + 6×0.25 + 9×0.15 = 2.80 + 1.75 + 1.50 + 1.35 = **7.40**. Y: 6×0.35 + 9×0.25 + 9×0.25 + 6×0.15 = 2.10 + 2.25 + 2.25 + 0.90 = **7.50**. Y wins narrowly on cost and delivery despite weaker quality. Overturning factor (any one): a single critical quality failure or compliance finding against Y; X's superior quality on a quality-critical programme; capacity risk at peak season; the closeness of the scores argues for trial orders to both before committing.
6. Spinning standard counts — push (stable demand, capital-heavy, forecast production efficient). Grey fabric — push (generic, holds the decoupling stock). Dyeing — pull (colour and finish are buyer-specific; dye only against orders). Garment sewing — pull (styles are order-specific; sew only confirmed orders).
